Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/21/2004
PI Name:	Ladd, Anthony J.C. Ph.D.		
Project Title:	Microgravity Driven Instabilities in Gas-Fluidized Beds		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Granular flows		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32611-6005	**Congressional District:**	6
Comments:	PI has completed submission (kw 11/30)		
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	04/01/2000	**End Date:**	11/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	note - end date changed from 11/30/2003		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2382		
Performance Goal No.:			
Performance Goal Text:			
	The primary difficulty in laboratory studies of gas-fluidized beds is that the buoyancy mismatch makes the gravitational force on the particles unavoidably large. These systems can only be fluidized by high-velocity gas flows where the dynamics is immediately very complicated, and the bed is unstable as soon as it is fluidized. Thus it is not known whether or not gas-fluidized beds are stable to uniform fluidization at sufficiently low Stokes number, as would be expected from purely fluid-dynamical considerations, or whether other forces, such as cohesion, play a destabilizing role. A further important question is whether or not there is a progression of instabilities in gas-fluidized beds similar to that observed in liquid-fluidized beds, or whether the effects of inter-particle collisions make a fundamental change to the unstable flow patterns. Only the bubbly phase has so far been observed in gas-fluidized beds, which is associated with higher Reynolds number flows in liquid-fluidized beds. These critical questions can not be addressed experimentally for gas-fluidized beds because stable fluidization can not be achieved in earthÃ?Â?s gravity. This task is		

Task Description:	composed of ground-based experiments on liquid-fluidized beds and numerical simulations of both liquid-fluidized and gas-fluidized suspensions. The numerical simulations will be used primarily to investigate the dynamics of gas-fluidized beds under microgravity conditions. The laboratory experiments will be used to visualize and characterize the particle motions in liquid-fluid beds, and to investigate possible imaging techniques for a microgravity experiment. Comparisons between simulation and experiment for liquid-fluidized systems will serve to validate the numerical models. The objectives of this work are to answer the following two questions: 1) Can a packed bed of particles be uniformly fluidized by a slow gas flow, if the gravitational weight of the particles is sufficiently small? 2) Does a gas-fluidized bed in a reduced gravity environment have a progression of different unstable flows with increasing gas velocity, similar to a liquid- fluidized bed, or does it immediately enter the bubbly regime observed in earth's gravity? An understanding of the dynamics of gas-fluidized beds raises interesting and important scientific issues: What is the role of inertia? What are the limits of stability? What are the characteristic patterns of the unstable flows? Commercial fluidized beds are typically gas-fluidized, with particle Reynolds numbers upwards of 10-100, and the bubbling instability frequently degrades their performance by reducing the efficiency of contact between particles and fluid. Thus an improved understanding of the fundamental physics of gas- fluidized beds could lead to significant advances in industrial processing of particulate materials. In addition gas-fluidized beds make a bridge between liquid-fluidized beds where hydrodynamic interactions are dominant and granular mechanics where they are ignored: both these systems are better understood than the gas-fluidized case.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Two journal publications were prepared during this year's no-cost extension.
Bibliography Type:	Description: (Last Updated: 03/15/2007)
Abstracts for Journals and Proceedings	N.-Q. Nguyen and A.J.C. Ladd. "Microstructure in a settling suspension of hard spheres" Physical Review E 63:050401 , May-2004